E-COMMERCE CUSTOMER SPEND PREDICTION

Predicting Total Spend from Customer Behavior & Demographics

A Machine Learning Project Report

1. Project Background & Objective

The goal of this project is to predict a customer's Total Spend on an e-commerce platform using their demographic profile (age, gender, city), membership tier, and behavioral signals (items purchased, average rating given, discount usage, and recency of last purchase). An accurate spend predictor allows a business to flag high-value customers early, personalize retention offers, and forecast revenue contribution at the individual level.

2. Deep Dive: Data Preprocessing & Cleaning

Raw Data Profile

- Shape: 350 rows × 11 columns.
- Features: Customer ID, Gender, Age, City, Membership Type, Total Spend (target), Items Purchased, Average Rating, Discount Applied, Days Since Last Purchase, Satisfaction Level.

Cleaning Operations Executed

Duplicate Check: 0 exact duplicate rows found — no deduplication required.

Missing Values: 2 records (Customer IDs 172 and 244, both from Houston) were missing a Satisfaction Level. Both were imputed with the column mode, “Satisfied.”

Outlier Screening (IQR): Total Spend and Days Since Last Purchase were checked against 1.5×IQR fences. Zero outliers were found in either column — the dataset's numeric ranges are naturally well-behaved, so no capping or trimming was needed.

3. Exploratory Data Analysis

Numerical features (Age, Total Spend, Items Purchased, Average Rating, Discount Applied, Days Since Last Purchase) and categorical features (Gender, City, Membership Type, Satisfaction Level) were examined through scatter and bar plots to surface early behavioral patterns:

- Younger customers (around age 27–28) were associated with both the highest ratings given and the highest total spend, suggesting an engaged, high-value younger segment.
- Total Spend rises with Items Purchased, reinforcing that purchase volume is a primary spend driver rather than price-per-item alone.
- San Francisco customers showed the highest average spend, while Houston customers showed the lowest — a clear regional gap worth investigating for market-specific pricing or promotions.

- Spend and Satisfaction Level appeared broadly balanced across age groups, without a single age band dominating dissatisfaction.

4. Feature Engineering & Encoding

Ordinal Encoding

- Membership Type: Gold → 0, Silver → 1, Bronze → 2 (reflecting descending tier value).
- Satisfaction Level: Unsatisfied → 0, Neutral → 1, Satisfied → 2.
- Discount Applied: Boolean True/False cast directly to 1/0.

One-Hot Encoding

Gender and City were expanded into indicator columns via one-hot encoding, allowing the linear model to assign an independent coefficient to each category level.

Standardization

Age, Items Purchased, Average Rating, and Days Since Last Purchase were scaled to zero mean and unit variance using StandardScaler prior to model fitting, putting all numeric predictors on a comparable footing.

5. Model Implementation & Results

A standard multiple Linear Regression model was fit on an 80/20 train-test split (random_state = 42) to predict Total Spend from the full engineered feature set.

Test R^2 Score: 0.9981 (99.8% of variance explained)

Metric Verification: As-Written vs. Corrected Pipeline

Two implementation details in the original notebook were reviewed for correctness before finalizing these results:

RMSE calculation: the notebook computed an extra, unintended square root on top of an already-computed RMSE value, printing 0.21 instead of the true error. The corrected RMSE, evaluated on real Total Spend units, is ₹15.92 — the model's typical prediction error is under ₹16 on spend values averaging ₹845, consistent with the very high R^2.

Scaling order: the notebook fit StandardScaler on the full dataset before the train/test split, which is a leakage risk in general. For this specific case the effect is negligible: ordinary Linear Regression's R^2 is mathematically invariant to feature scaling, so predictions are unaffected — confirmed by rerunning the pipeline with the scaler fit strictly on the training set only. The table below shows both versions for transparency.

Metric	As-Written Pipeline	Corrected (Leakage-Free) Pipeline
R^2 Score	0.9981	0.9981
RMSE	₹15.92	₹15.89
MAE	₹11.35	₹11.33

Feature Coefficients (Corrected Pipeline, Real Currency Units)

Coefficients below reflect the corrected, leakage-free pipeline (scaler fit on training data only), expressed directly in currency units for interpretability. Sorted by absolute impact on predicted Total Spend:

Feature	Coefficient (₹ per unit)	Direction
Membership Type (Gold→Bronze)	-243.35	↓ Lower tier
City: San Francisco	+152.21	↑ Higher
Items Purchased	+100.97	↑ Higher
City: Miami	-93.01	↓ Lower
City: Los Angeles	-59.70	↓ Lower
City: Houston	-51.30	↓ Lower
City: New York	+45.82	↑ Higher
Discount Applied	-41.21	↓ Lower
Days Since Last Purchase	-19.14	↓ Lower
Satisfaction Level	-9.10	↓ Lower

6. In-Depth Discussion & Findings

1. Membership Type Is the Single Strongest Driver

Finding: Membership Type carries the largest coefficient by a wide margin (-243.35 per tier step from Gold toward Bronze), meaning membership tier alone accounts for a swing of several hundred rupees in predicted spend.

Interpretation: This is expected and desirable — membership tiers are typically assigned based on historical spend, so the model is correctly picking up on a strong, pre-existing business signal rather than an unexplained artifact.

2. Geography Is a Secondary but Consistent Signal

Finding: San Francisco (+152.21) and Miami (-93.01) sit at opposite ends of the city coefficients, matching the EDA observation that San Francisco customers spend the most and Houston customers the least.

Interpretation: Regional spend gaps this consistent across both descriptive EDA and model coefficients suggest genuine market-level differences worth reflecting in region-specific marketing budgets.

3. Near-Perfect R^2 on a Small, Clean Dataset

Finding: At $R^2 = 0.9981$ on only 350 records (280 training / 70 test), the linear relationship between these features and Total Spend is close to deterministic.

Caution: With a test set of just 70 customers, this R^2 should be treated as an encouraging signal rather than a guarantee — k-fold cross-validation across multiple splits would give a more robust confidence interval before this model is used for business decisions.

4. Minor Cleanup Opportunity

Customer ID was left in the feature set. Its coefficient (₹0.017) is negligible and does not materially affect predictions here, but as a non-informative identifier it should be dropped in any future iteration to keep the feature set strictly meaningful.

5. Takeaway

A simple, fully interpretable Linear Regression model explains over 99% of the variance in customer spend using only demographic and behavioral features that any e-commerce platform already collects — with membership tier and purchase volume as the clearest levers for targeted retention and upsell strategies.

Prepared as part of an academic machine learning project.